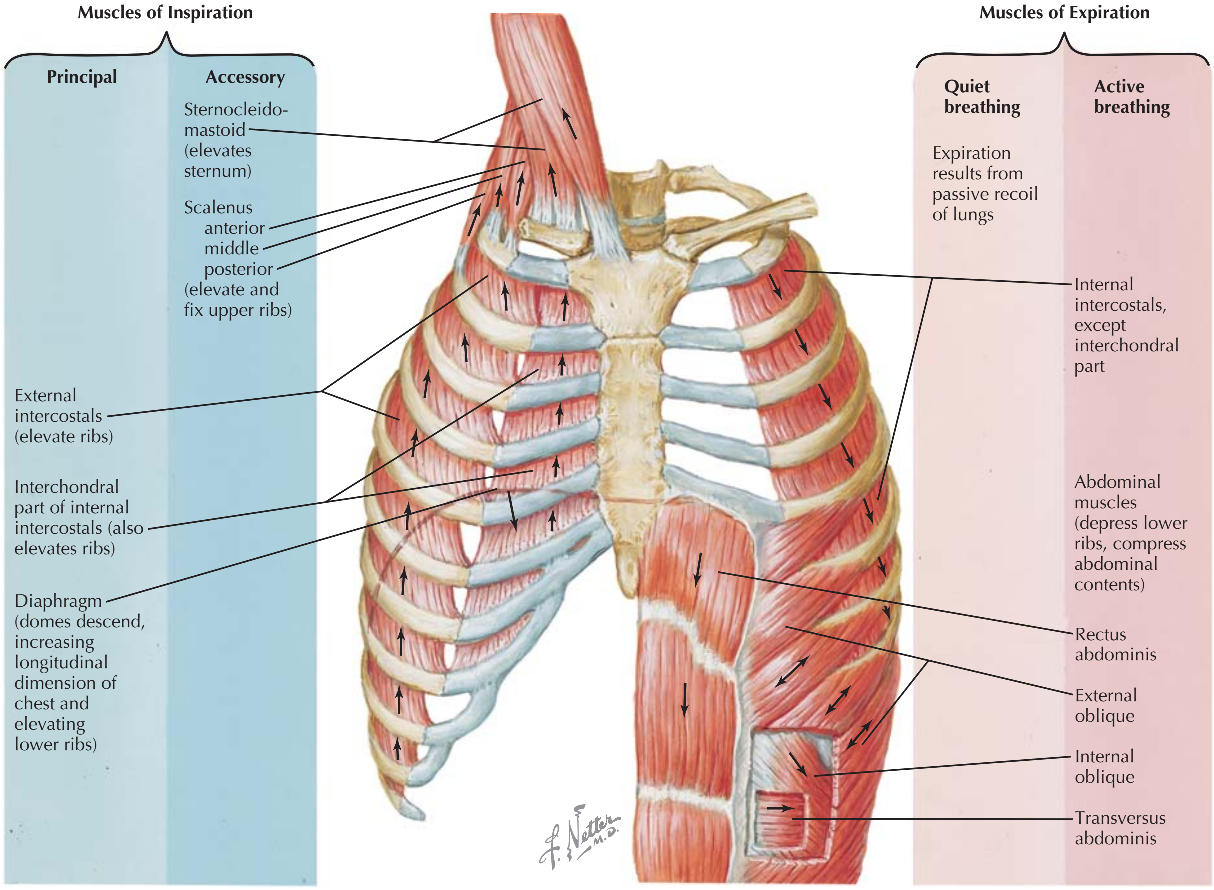


Figure 14.1 Respiratory Muscles Contraction of the diaphragm is the main factor producing inspiration during normal, quiet breathing; expiration is a passive process in this type of breathing, caused by passive recoil of the lungs. Active breathing requires the activity of additional muscles and involves energy expenditure for both inspiration and expiration.

recoil pressure is still negative, but recoil pressure of the total respiratory system is positive due to increased recoil pressure of the lungs (see Fig. 14.3). At total lung capacity, both chest wall and lung recoil pressures are positive.

During **expiration,** with the relaxation of inspiratory muscles, the elastic recoil pressure of the respiratory system (increased due to its expansion) results in elevation of alveolar pressure above atmospheric pressure, causing outward airflow until atmospheric pressure is reached in the lungs (see Fig. 14.2*C*). The system returns to FRC, unless air is actively expired beyond that level; expiration is eventually limited by the large negative elastic recoil pressure of the chest wall as residual volume (RV) is approached.

> During normal, quiet breathing, inspiration is mainly driven by the contraction of the muscular diaphragm, whereas expiration is a passive process, driven by the positive elastic recoil pressure of the respiratory system achieved during inspiration. In this type of breathing, pleural pressure is always negative. During active expiration, contraction of abdominal muscles compresses the viscera, forcing the diaphragm upward, producing positive pleural pressure. Thus, in contrast to quiet breathing, in which work is performed only for inspiration, energy is expended for both inspiration and expiration in active breathing.

COMPLIANCE, ELASTANCE, AND PRESSURE–VOLUME RELATIONSHIPS

Elastance is defined as the tendency of a hollow organ to return to its original size when distended; it can be quantified as elastic recoil pressure. The **elastic recoil** of the lungs and chest wall can be related to pleural pressure:

- Elastic recoil of the lungs is equal to alveolar pressure minus pleural pressure.
- Elastic recoil of the chest wall is equal to pleural pressure minus atmospheric pressure.

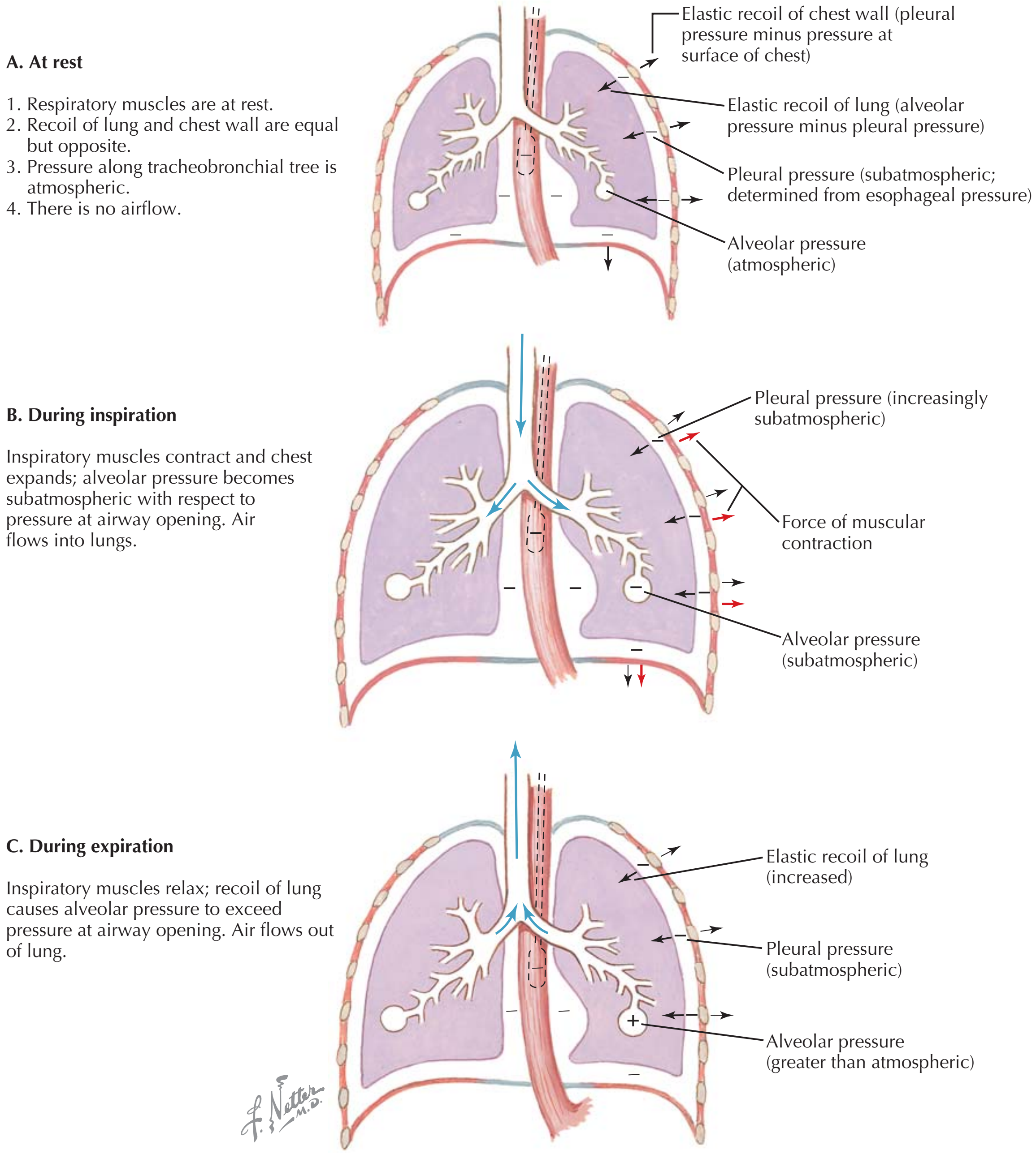


Figure 14.2 **Forces during Quiet Breathing** Normal, quiet breathing is produced mainly by contraction of the diaphragm, resulting in inspiration as alveolar pressure falls below atmospheric pressure; expiration occurs when the diaphragm relaxes and recoil pressure of the lungs elevates alveolar pressure above atmospheric pressure. The dynamic interactions of elastic recoil pressure of the lungs and chest wall and contraction and relaxation of the diaphragm producing airflow are illustrated *(A–C).* Pleural pressure is always negative during quiet breathing.

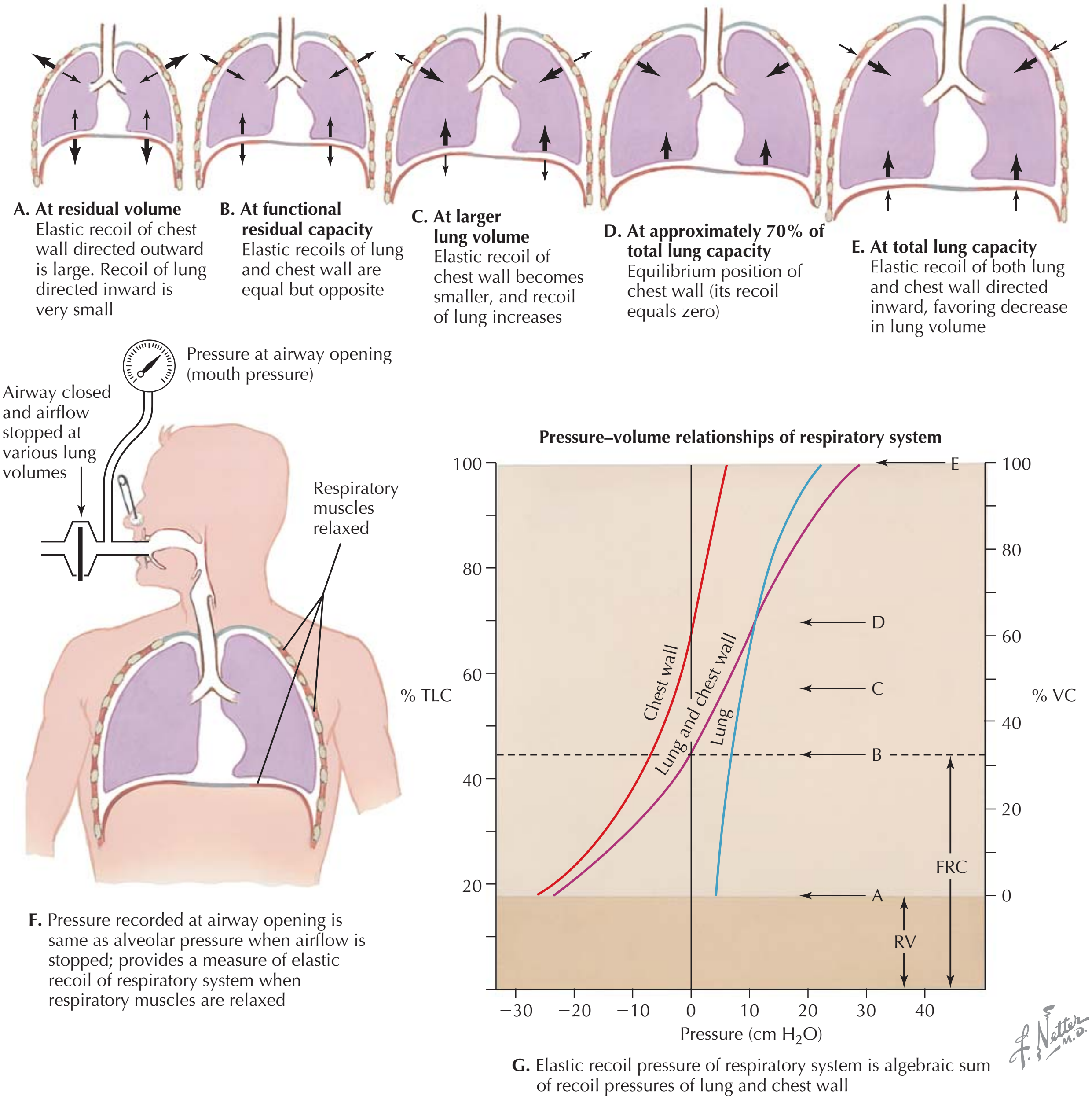
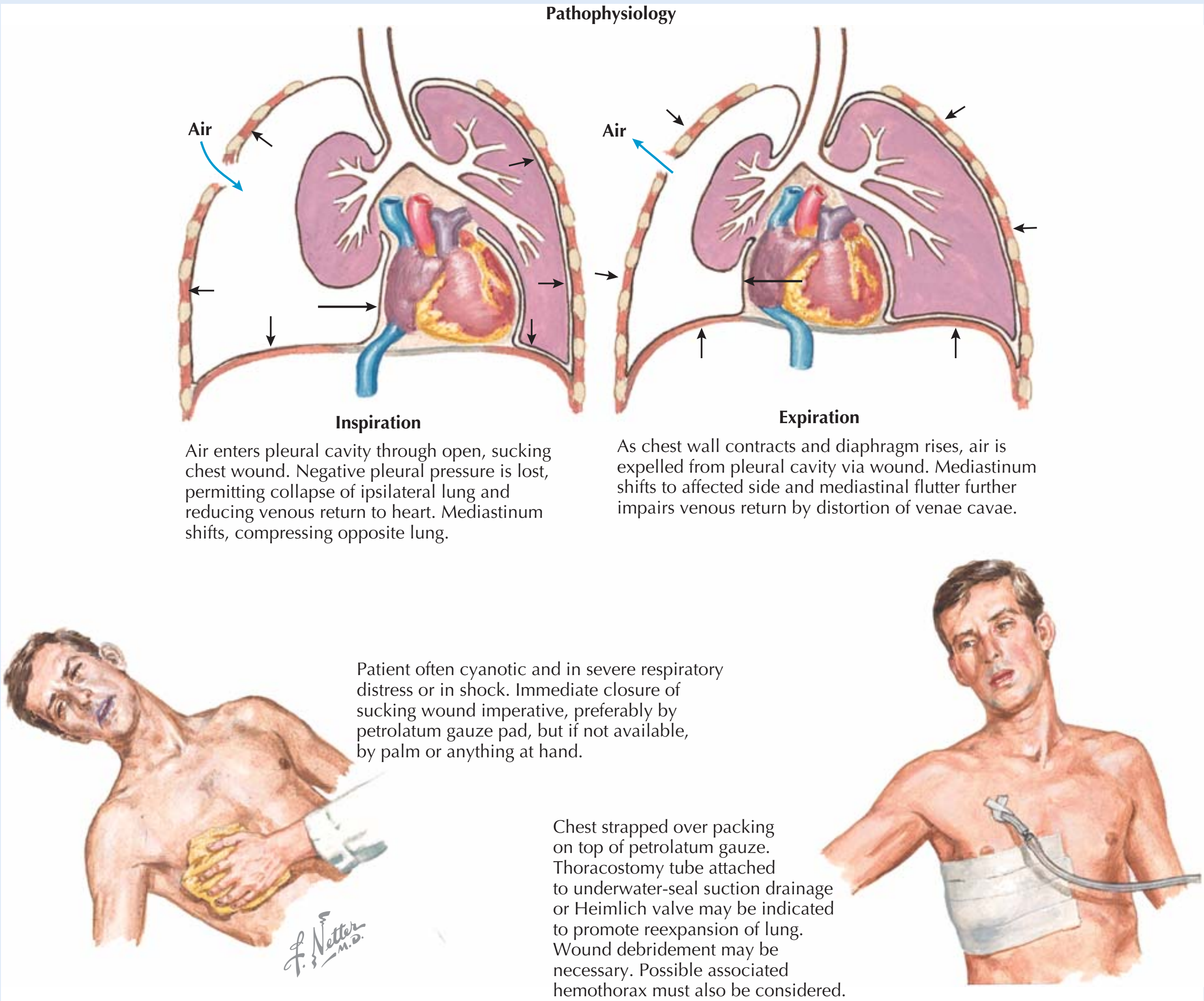


Figure 14.3 **Elastic Properties of Respiratory System: Lung and Chest Wall** Elastic recoil pressure of the respiratory system is the algebraic sum of recoil pressures of the lungs and chest wall *(G).* These pressures can be measured at various lung volumes from residual volume to total lung capacity *(A–E),* and represent the forces resulting in inspiration and expiration. Above functional residual volume (FRC), the net recoil pressure of the respiratory system is positive, and relaxation of inspiratory muscles will result in expiration; above FRC, net recoil pressure is negative and relaxation of expiratory muscles will cause inspiration. At FRC, the system is in equilibrium. Elastic recoil when respiratory muscles are relaxed can be determined by stopping airflow *(F).* RV, residual volume.

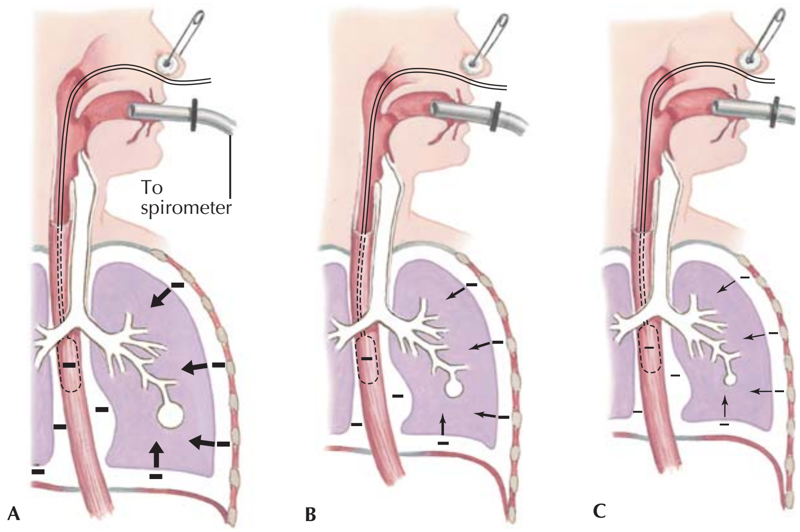
Lung compliance (CL) is a measure of the distensibility of the lung and is the inverse of lung elastance. Thus, CL is measured as the change in lung volume resulting from a change in transpulmonary pressure, where transpulmonary pressure is the pressure across the lung, or the difference between alveolar and pleural pressure (Fig. 14.4). Experimentally, compliance can be measured in an isolated lung (Fig. 14.5). Referring to the experimental preparation in Figure 14.5 in which the lung is filled with air, as pressure applied to the lung is changed, the volume of the lung changes. The slope of this curve is the compliance of the lung. Note that the slope differs between inflation and deflation, with the volume at a given pressure being lower during inflation than during deflation, a phenomenon known as **hysteresis.** The difference in the inflation and

CLINICAL CORRELATE
Pneumothorax

Pneumothorax is a condition in which air accumulates in the pleural cavity, due to injury to the chest wall or lungs or as a result of pulmonary disease. Ordinarily, the pleural cavity contains only a few milliliters of fluid. Introduction of air into the pleural space through an open (sucking) pneumothorax will uncouple the mechanical forces associated with the chest walls and lung, causing collapse of the lung on the damaged side, thereby affecting ventilation in that lung. In a tension pneumothorax, air is able to enter the pleural space but cannot leave (a piece of tissue acts as a one-way valve). Air accumulates with each breath, raising intrathoracic pressure and thereby causing severe dyspnea (shortness of breath) and circulatory collapse. Tension pneumothorax is an unstable condition requiring immediate medical intervention.



Open Pneumothorax



During a slow expiration from TLC, flow is periodically interrupted and measurements are made of lung volume and of transpulmonary pressure. Transpulmonary pressure is the difference between alveolar and pleural pressures. Pleural pressure is determined from pressure in the esophagus. Because there is no airflow, alveolar pressure is the same as pressure at the airway opening.

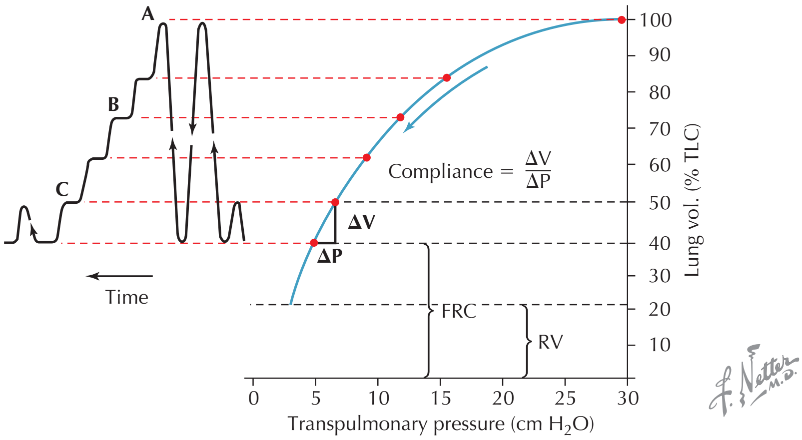


Figure 14.4 **Measurement of Elastic Properties of the Lung** Lung compliance (ΔV/ΔP) is a measure of lung distensibility and can be measured by determining transpulmonary pressure at various lung volumes *(A, B,* and *C).* Transpulmonary pressure is the difference between alveolar and pleural pressure. Alveolar pressure is the same as pressure measured at the airway opening when flow is stopped; pleural pressure is measured by esophageal balloon catheter. Note that lung compliance is highest at residual volume, and is reduced at greater lung volumes. TLC, total lung capacity.

deflation curves is mainly associated with the effects of surface tension during inflation, when surface tension at the liquid-air interface of the lung must be overcome as volume rises. When a fluid-filled lung (in which surface tension is not an issue) is exposed to the same manipulations, compliance is greater and hysteresis is less pronounced.

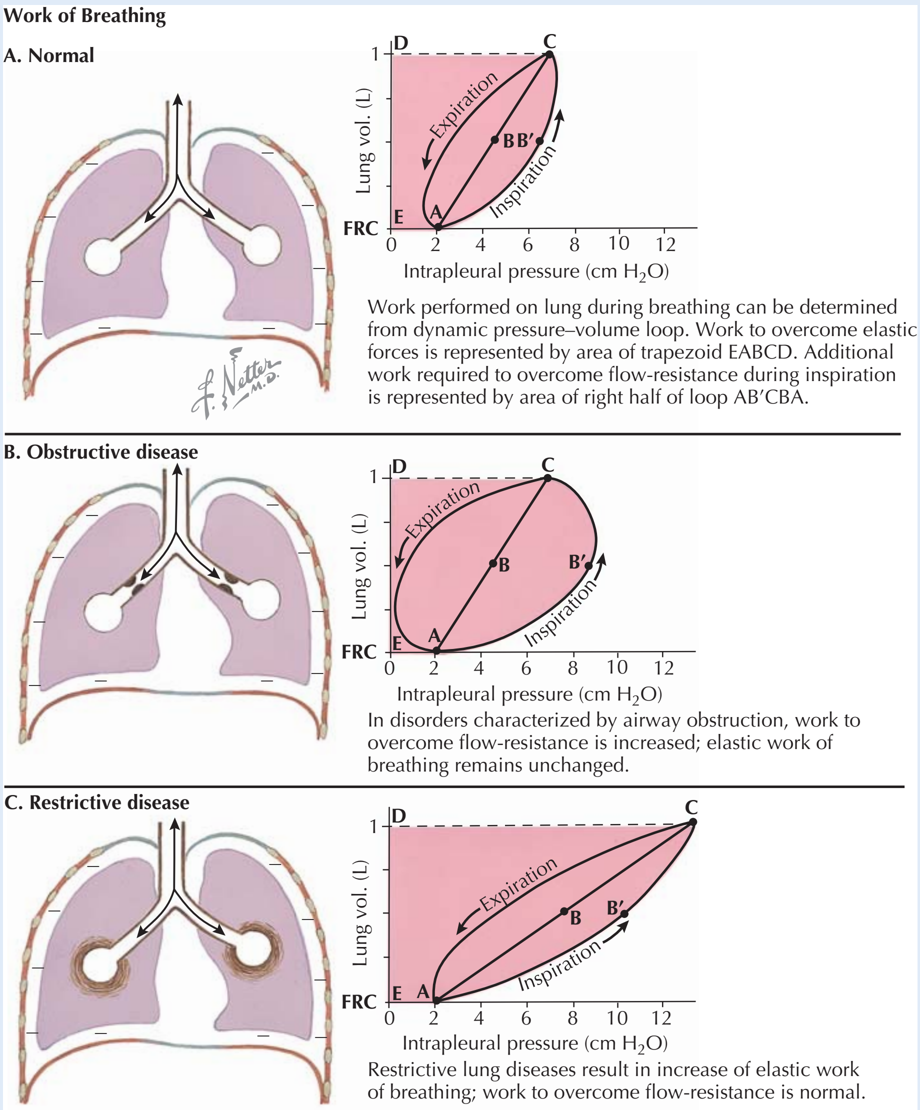
Like the lungs, the chest wall also is characterized by compliance and elastance. These properties become apparent when the chest wall is punctured, creating a **pneumothorax** (see “Pneumothorax” Clinical Correlate).

SURFACTANT AND SURFACE TENSION

Surface tension is the elastic-like force at the surface of a liquid (at the gas-liquid interface) caused by intermolecular attraction of the liquid molecules at that surface. In the lung, surface tension reduces lung compliance and has the potential for causing collapse of small airways. The potential problems of surface tension and low pulmonary compliance are overcome by the production of **surfactant** by type II alveolar epithelial cells. Surfactant is a complex lipoprotein containing the phospholipid **dipalmitoyl phosphatidyl choline.** It is

CLINICAL CORRELATE
Compliance in Pulmonary Diseases

Lung diseases are often classified as restrictive and obstructive diseases. Restrictive diseases are characterized by reduced functional volume of the lung, whereas obstructive diseases are characterized by reduced flow rate. Restrictive diseases include interstitial lung diseases (for example, idiopathic pulmonary fibrosis, sarcoidosis, and asbestosis); examples of obstructive lung diseases include chronic obstructive pulmonary disease (COPD) and asthma. In restrictive diseases, compliance of the respiratory system is reduced, resulting in lower FRC and total lung capacity, and reduced slope of the pressure–volume relationship. Obstructive lung disease does not directly affect lung compliance but may ultimately change compliances. For example, in emphysema, often associated with tobacco smoking and COPD, elastic fibers in the lungs are destroyed, alveolar architecture is compromised, and lung compliance is increased (elasticity is reduced). FRC and total lung capacity are increased, and the slope of the pressure–volume relationship is greater. Despite increased compliance and greater tidal volume, emphysema sufferers have reduced ability to exchange gas, based on the destruction of alveoli.



Work of Breathing in Obstructive and Restrictive Lung Disease Compared to work performed in normal breathing *(A),* in obstructive disease, while compliance may be unchanged, the work of breathing is increased by the elevated airway resistance *(B).* In restrictive disease, lung compliance is low, and the elastic work of breathing is increased *(C).*

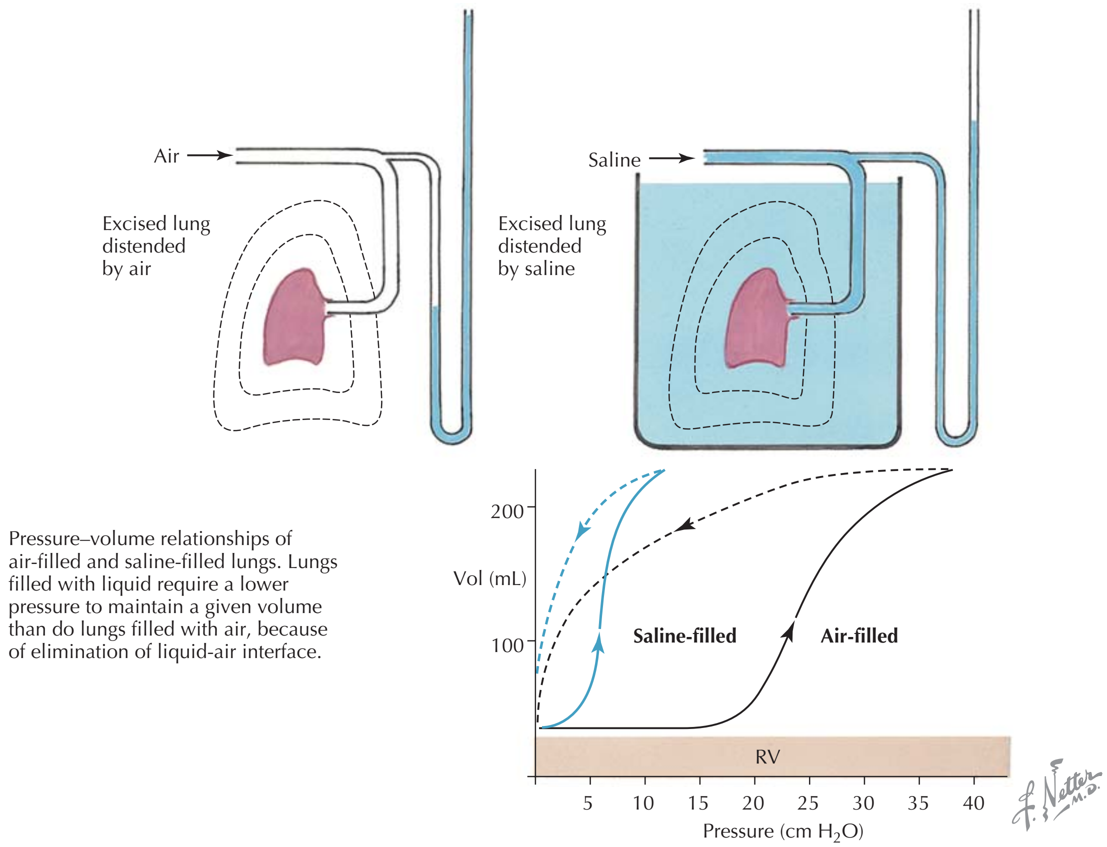


Figure 14.5 **Compliance and Surface Forces of the Lung** Measurement of the pressure–volume relationship in an isolated lung results in the illustrated pressure–volume loops. Greater pressure is required to inflate an air-filled lung compared with a saline-filled lung, due to the surface tension at the air-alveolar interface in the former. Surfactant is produced by type II alveolar epithelial cells and reduces surface tension of the air-filled lung; without it, even greater force would be required to inflate the lungs with air. RV, residual volume.

amphipathic and lines the surface of the alveolar epithelium and small airways (with hydrophilic regions of the phospholipid oriented toward the epithelial surface and hydrophobic regions facing the lumen). Surfactant reduces surface tension of airways and alveoli and increases lung compliance, reducing the work of breathing.

AIRWAY RESISTANCE

Flow of air in and out of the lungs is dependent on the pressure gradient from the opening of the mouth to the alveoli. At the end of inspiration or expiration, the gradient is zero. **Poiseuille's law,** presented in the context of blood flow in Section 3, also applies to flow of air (Q) through tubes (see Fig. 14.5):

$$Q = \frac{\Delta P \pi r^4}{\eta 8L}$$

where ΔP is the pressure gradient from one end of a tube to the other, r^4 is the radius of the tube to the fourth power, η is the viscosity of air, and L is the length of the tube. Thus, based on this relationship, airflow (Q) through a tube should be:

- Directly proportional to the longitudinal pressure gradient (inflow pressure minus outflow pressure).
- Inversely proportional to the length of the tube.
- Inversely proportional to the viscosity of air.
- Directly proportional to the fourth power of the radius of the tube.

In the respiratory system as a whole, the *greatest resistance to flow actually occurs in medium-sized airways* (fourth to eighth generation). Recall from Section 3 that in parallel tubes, total resistance is less than the resistance of the individual tubes. Considering both the diameter and number of tubes at this level, resistance is higher in the medium-sized bronchi (in aggregate) than in the larger or smaller airways.

The effects of surface tension in the lung have often been incorrectly presented in textbooks in the context of **Laplace's law,** which states that in a sphere, surface tension creates pressure (P):

$$P = \frac{2T}{r}$$

where T is surface tension and r is radius of the sphere. According to this construct, in an isolated alveolus, this pressure is the pressure tending to cause its collapse (or the equivalent pressure required to keep the alveolus open). Laplace's law would lead to the prediction that when multiple spherical units are subjected to the same inflation pressure through a branching tube, larger units would expand and smaller units would collapse, as the higher surface tension in smaller units causes air to flow toward larger units in which surface tension is lower. Textbooks have typically illustrated this situation as a Y-tube with two alveoli attached, or as alveoli in the configuration of a "bunch of grapes." According to this line of argument, a major role of surfactant is to overcome the effects of surface tension in alveoli of different sizes and promote more uniform inflation of the lungs.

This argument is incorrect for several reasons: alveoli are actually prismatic in shape (appearing polygonal in histological sections) and exist in a honeycomb-like geometry with common walls, sometimes even with pores between them. Thus, alveoli do not act as independent units, and inflation of an alveolus affects inflation of adjacent alveoli. Rather than viewing surfactant's role in terms of alveolar surface tension, its effects on the overall compliance of the lung and on small airway patency should be emphasized.

(From Prange H: Laplace's Law and the alveolus: A misconception of anatomy and a misapplication of physics. Advan Physio Edu 27:34–40, 2003.)

Poiseuille's law applies to laminar flow of air but becomes less precise with turbulent flow. Laminar flow occurs in small airways, whereas flow in the largest airways is turbulent; flow in the remainder of the system tends to be transitional, with some turbulence (Fig. 14.6). Factors producing **turbulent and laminar flow** are discussed in Section 3, in the context of blood flow. In the largest airways, high velocity and airway diameter contribute to producing turbulent flow. In small, peripheral airways, smaller diameter and lower velocity result in laminar flow of air.

Effects of Autonomic Nerves on Airway Resistance

Airway resistance is also affected by the autonomic nervous system:

- **Activation of parasympathetic nerves** innervating smooth muscle of conducting airways causes bronchoconstriction and promotes glandular secretions in the lungs.
- **Activation of sympathetic nerves** innervating smooth muscle of conducting airways results in bronchodilation and reduced airway resistance (through activation of β_2-adrenergic receptor-linked pathways) in mammalian species, although human lungs have little sympathetic innervation. Release of epinephrine by the adrenal medulla during sympathetic activation will also reduce airway resistance through activation of the pulmonary β_2-receptor mechanism.

CLINICAL CORRELATE

Treatment of Asthma with Sympathetic Adrenergic Drugs

Asthma is a chronic respiratory disease, characterized by episodic bronchoconstriction and mucus secretion, resulting in increased airway resistance and dyspnea (labored breathing). Asthma can be life-threatening in severe cases. Asthma attacks may be initiated by a variety of factors, including inhaled allergens and irritants, cold air, stress, and exercise, which exacerbate the ongoing airway inflammation in asthmatics. Drugs used to prevent asthma attacks are mainly aimed at reducing inflammation, whereas drugs used to relieve acute attacks are generally bronchodilators (e.g., inhaled β_2-adrenergic agonists such as salbutamol or terbutaline). These drugs reduce airway resistance by relaxing bronchial smooth muscle.

Lung Volume and Airway Resistance

In addition, there is a relationship between **lung volume and airway resistance.** At high lung volume, the diameter of airways tends to be increased by the radial traction the expanded lungs place on the airways. At low lung volumes, in the absence of traction, small airways have a greater tendency to collapse.

DYNAMIC COMPRESSION OF AIRWAYS DURING EXPIRATION

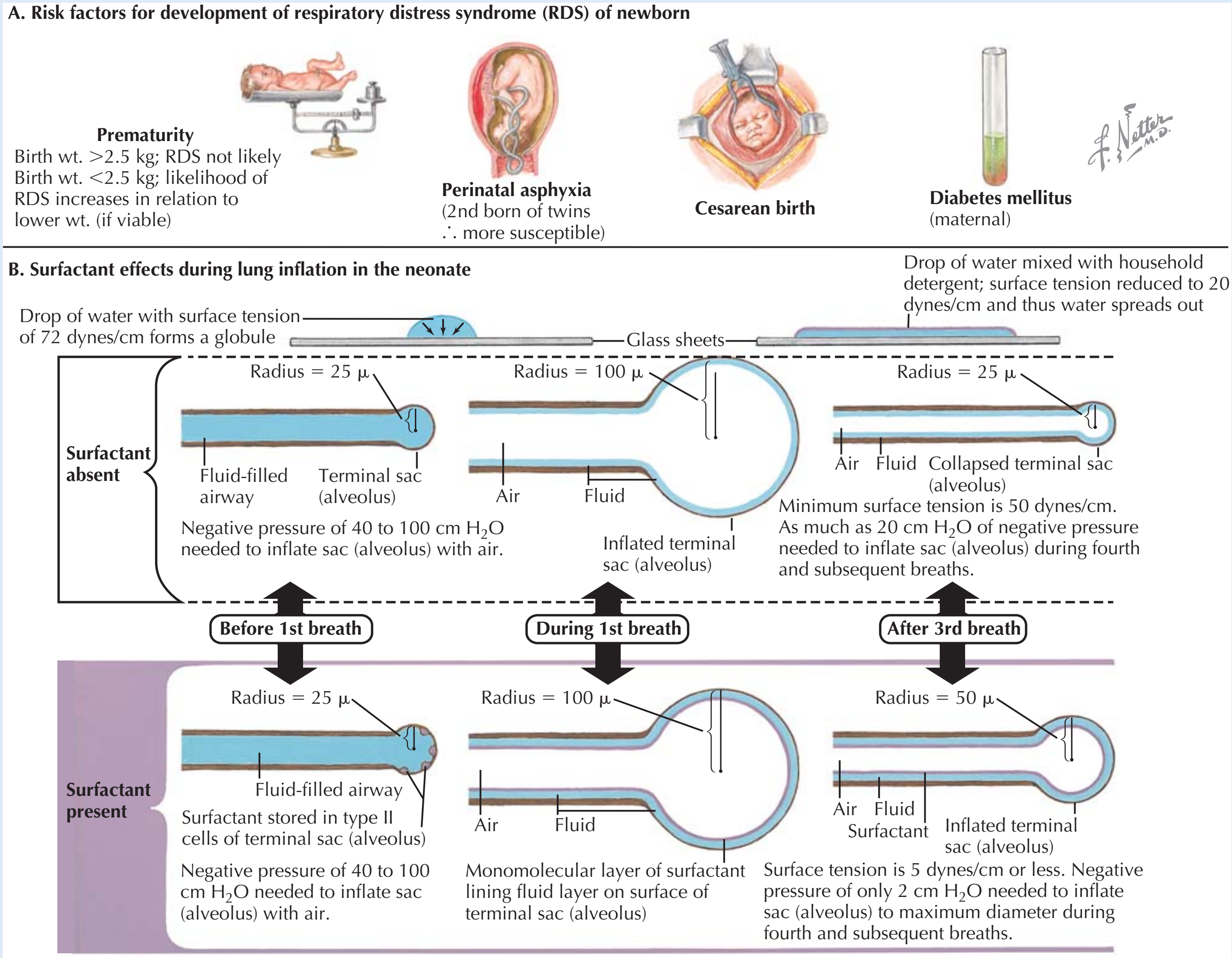
Airway resistance is also affected by **dynamic compression,** which is the compression of airways during *forced expiration*. An **expiratory flow–volume curve** is illustrated in Figure 14.7 (*A*, solid line), in which a subject performs a **forced vital capacity maneuver,** inspiring to total lung capacity and then exhaling as forcibly as possible to residual volume. Note that in the expiratory flow–volume curve generated:

- The peak of this curve represents the **peak expiratory flow rate (PEFR).**
- The downward slope (expiratory phase) of the flow-volume curve is **effort-independent;** during this phase of the curve, flow is limited by **dynamic compression** of the airways.

When less effort is exerted, lower peak flow is reached, but the associated flow–volume curve converges with the maximum effort curve and the downslope is the same, consistent with the effort-independence of expiratory flow during active expiration (see Fig. 14.7*A, dotted line*).

CLINICAL CORRELATE
Respiratory Distress Syndrome of the Newborn

Immediately after delivery, a newborn takes its first breath. Negative pressure of 40 to 100 cm H_2O is required to draw air into the collapsed airways and inflate the alveoli. During this first breath, in a healthy, normal, term infant, surfactant stored in type II alveolar epithelial cells is released and forms a mononuclear layer at the air-fluid interface of the small airways and alveoli. By the third breath, only a small negative pressure is required to inflate the lung. Respiratory distress syndrome (previously known as hyaline membrane disease) is the most common cause of death in premature infants, and is caused by a lack of surfactant production. Without surfactant, the negative pressure required to inflate the lungs remains high (CL is low) and portions of the lung collapse, leading to respiratory distress and potentially respiratory failure and death. Treatment consists of ventilatory support and surfactant therapy through the breathing tube.



The initial inflation of the collapsed lungs of a neonate requires large negative pressure (40 to 100 cm H_20), but surface tension is reduced in subsequent breaths as surfactant lines the alveoli and small airways, reducing the work required to inflate the lungs. Premature birth is often associated with surfactant deficiency and respiratory distress.

During normal, quiet breathing, pleural pressure is always negative. However, during active expiration, contraction of expiratory muscles results in elevation of pleural pressure above atmospheric pressure (see Fig. 14.7). Under these circumstances, during expiration, alveolar pressure is the sum of the positive pleural pressure and the elastic recoil pressure of the lungs. Pressure in the airways falls between the alveoli and the opening of the mouth, reaching atmospheric pressure at the mouth. Thus, at some point downstream from the alveoli, an **equal pressure point** is reached, at which airway

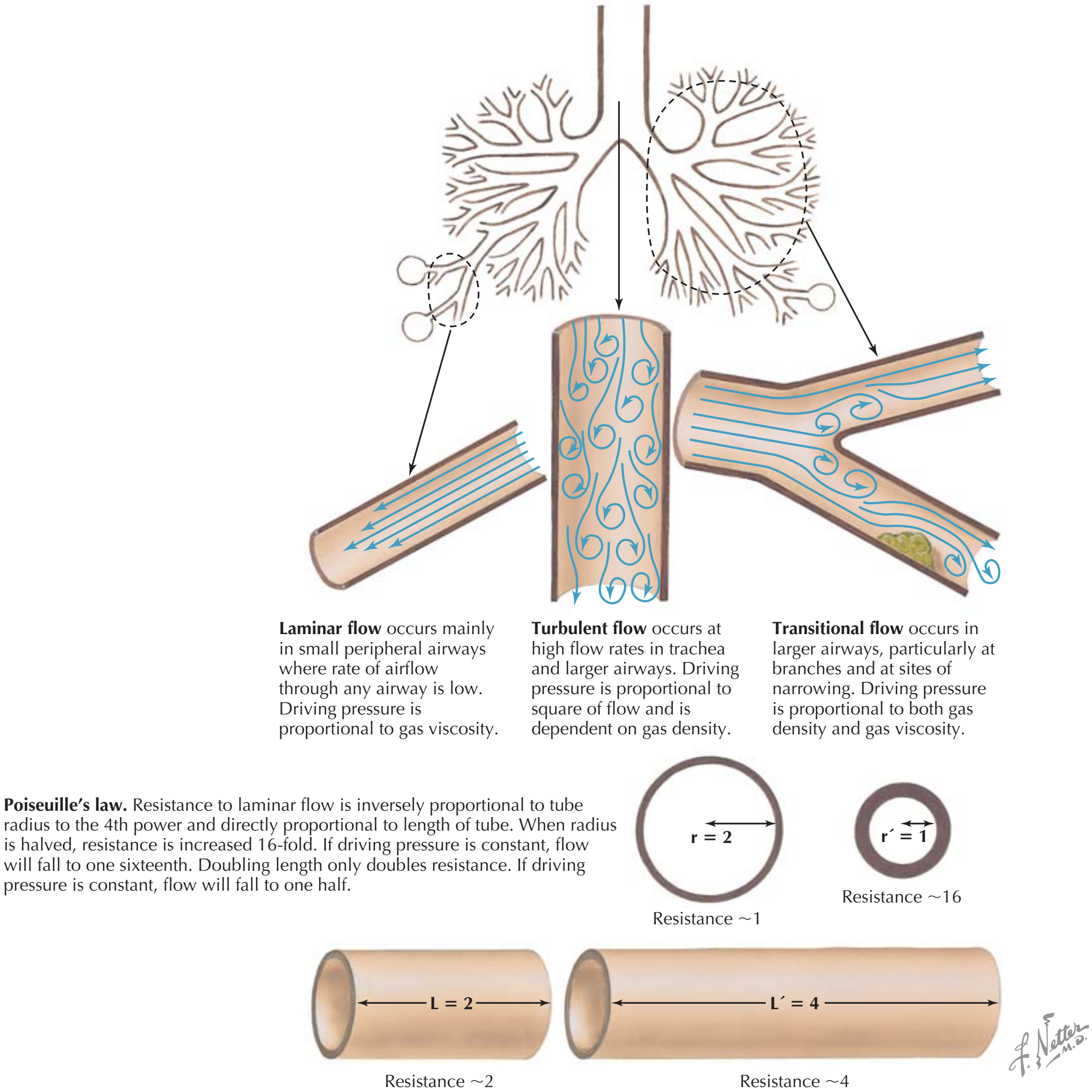
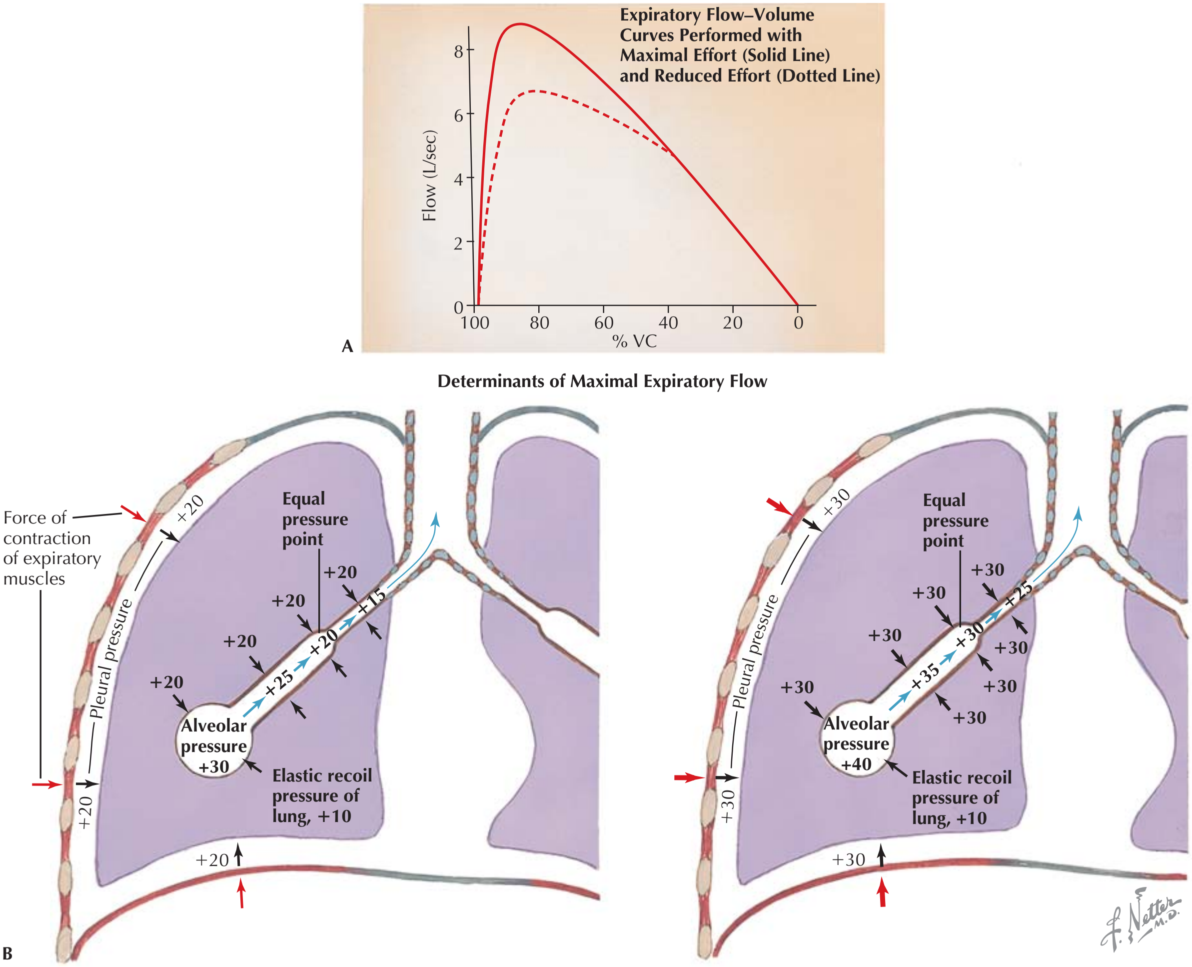


Figure 14.6 **Flow** Airflow through large airways tends to be turbulent, whereas flow in smaller airways tends to be laminar. Laminar flow is described by Poiseuille's law. Airway diameter, according to this law, is the greatest factor in determining resistance and flow (flow is proportional to the fourth power of the radius of a tube).

pressure is equal to pleural pressure. Beyond this point, airways will be compressed. This **dynamic compression** of airways limits expiratory flow rate and accounts for the effort-independent nature of airflow in active expiration. When greater effort is exerted, greater compression occurs, so that airflow remains the same.

OBSTRUCTIVE AND RESTRICTIVE PULMONARY DISEASES AND PULMONARY FUNCTION TESTS

Measurements of flow–volume relationships are important in assessing **obstructive and restrictive pulmonary diseases** (see "Compliance in Pulmonary Diseases" Clinical Correlate). Severe chronic obstructive pulmonary disease (COPD) is characterized by emphysema, in which inflammation leads to destruction of alveolar walls and capillaries, resulting in high lung compliance. Reduced elastic recoil of the alveoli and airways also results in early formation of the equal pressure point (at a point closer to the alveolus) during expiration, and, as a result of this dynamic compression, "trapping" of air takes place in the lung. These changes ultimately produce an increase in total lung capacity (TLC), functional residual capacity (FRC), and residual volume (RV). The changes in flow are clinically assessed by spirometry and related tests (Fig. 14.8). Notably, expiratory flow rate and **FEV$_1$** (the forced



At onset of maximal airflow, contraction of expiratory muscles at a given lung volume raises pleural pressure above atmospheric level (+20 cm H_2O). Alveolar pressure (sum of pleural pressure and lung recoil pressure) is yet higher (+30 cm H_2O). Airway pressure falls progressively from alveolus to airway opening in overcoming resistance. At equal pressure point of airway, pressure within airway equals pressure surrounding it (pleural pressure). Beyond this point, as intraluminal pressure drops further, below pleural pressure, airway will be compressed.

With further increases in expiratory effort, at same lung volume, pleural pressure is greater and alveolar pressure is correspondingly higher. Fall in airway pressure and location of equal pressure point are unchanged, but beyond equal pressure point, intrathoracic airways will be compressed to a greater degree by higher pleural pressure. Once maximal airflow is achieved, further increases in pleural pressure produce proportional increases in resistance of segment downstream from equal pressure point, so rate of airflow does not change.

Figure 14.7 Expiratory Flow–Volume Relationship The expiratory flow–volume curve generated during a forced vital capacity maneuver *(solid line in graph)* is characterized by a peak expiratory flow rate (PEFR) at the top of the curve and a downward slope during the remainder of expiration *(A)*. The maximal expiratory flow rate at various lung volumes along this downward slope is effort-independent, due to dynamic compression of airways *(B)*. When effort is reduced *(dotted line in graph)*, PEFR is lower, but note the overlap in the downward slope of the two lines, consistent with the effort-independence of maximal expiratory flow rate.

expiratory volume in the first second of a vital capacity maneuver) and the **FEV_1-to-FVC ratio** is less than the normal value of 75% (although FEV_1 is reduced, FVC is only slightly lower than normal). In contrast, in restrictive diseases such as interstitial fibrosis, thickening of alveolar walls results in decreased lung compliance. Lung volumes are reduced as a result (Fig. 14.9). Although FEV_1 is reduced, the FEV_1-to-FVC ratio is usually normal or elevated, because FVC is also diminished in restrictive disease. Various **pulmonary function tests** and typical findings in obstructive and restrictive disease are summarized in Figure 14.10.

Because of **dynamic compression** of airways, maximum expiratory flow is effort-independent. For any given lung volume, the associated maximum expiratory flow cannot be exceeded by increasing expiratory effort. The explanation for this phenomenon is that increased expiratory effort physically compresses airways as well as alveoli, increasing resistance and limiting airflow.

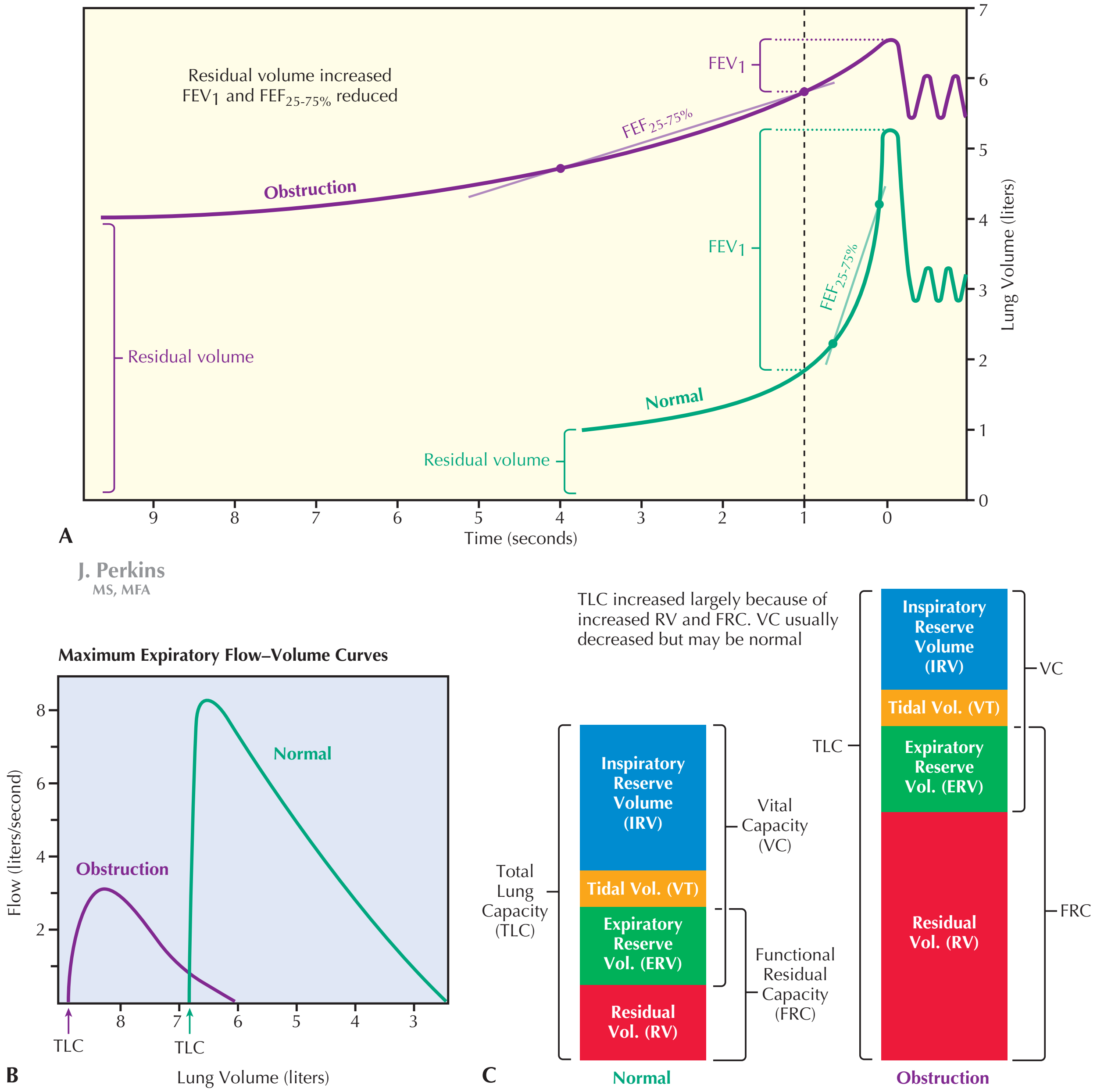


Figure 14.8 **Pulmonary Function in Obstructive Lung Disease** In emphysema, a chronic obstructive lung disease often associated with smoking, there is inflammatory destruction of elastic tissues in the lung, resulting in reduced elastic recoil of the lung. Changes in lung volumes *(A)*, flow–volume curves *(B)*, and spirometric measurements *(C)* associated with emphysema are illustrated. Notably, FEV_1 is reduced in obstructive lung disease, as is the ratio of FEV_1 to FVC *(A)*. $FEF_{25–75\%}$, the expiratory flow rate during the middle portion of a forced expiration, is also reduced. FEF, forced expiratory flow rate; FEV, forced expiratory volume.

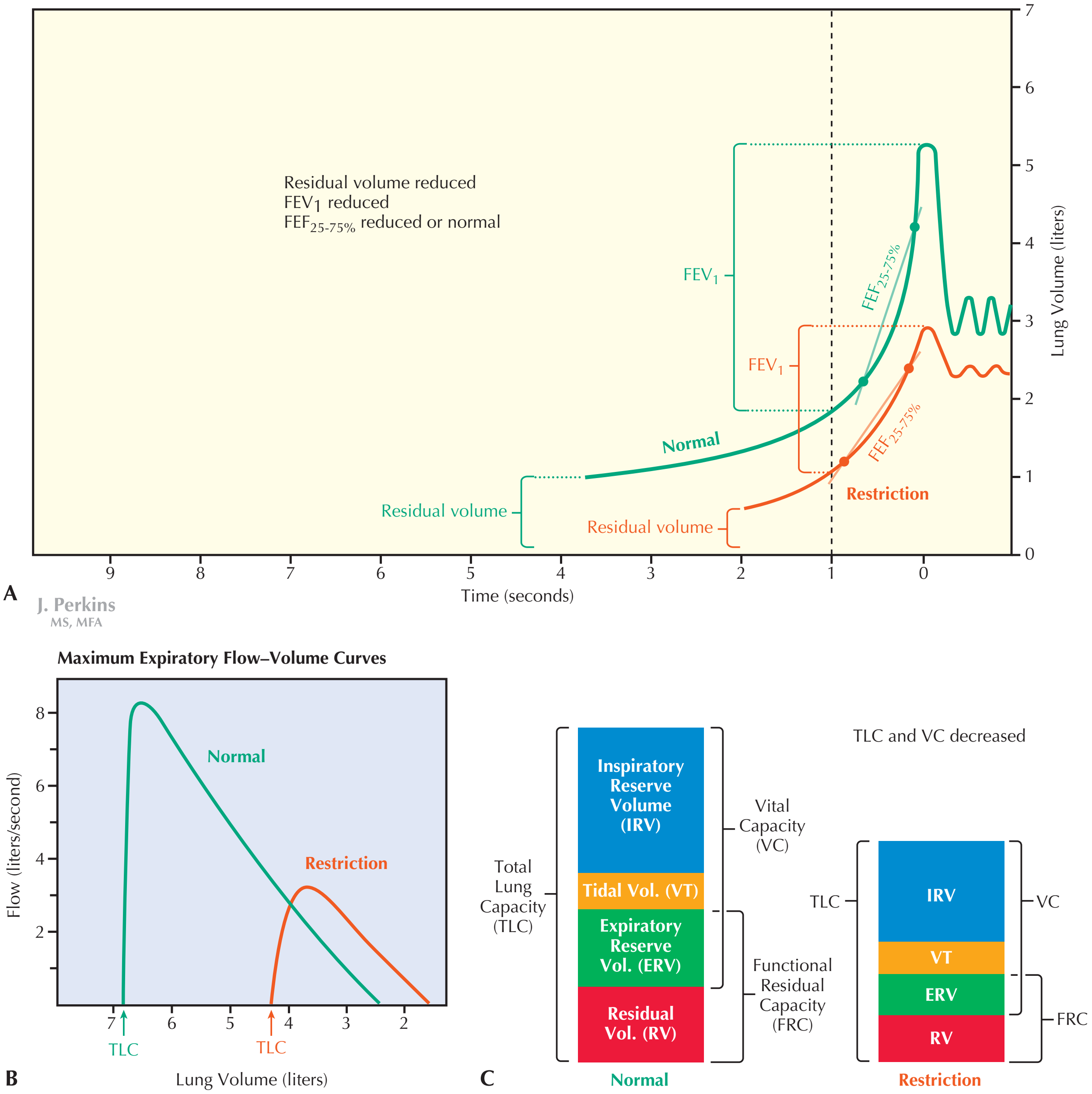


Figure 14.9 Pulmonary Function in Restrictive Lung Disease Lung compliance is reduced in restrictive lung diseases such as interstitial fibrosis, resulting in diminished lung volumes. Changes in spirometric measurements *(A),* flow–volume curves *(B),* and lung volumes *(C)* associated with restrictive lung disease are illustrated. Because both FEV_1 and FVC are reduced in restrictive lung disease *(A),* the ratio of FEV_1 to FVC is usually normal but may even be increased when FVC is greatly reduced. $FEF_{25–75\%}$, the expiratory flow rate during the middle portion of a forced expiration, is normal or reduced in restrictive disease. FEF, forced expiratory flow rate; FEV, forced expiratory volume.

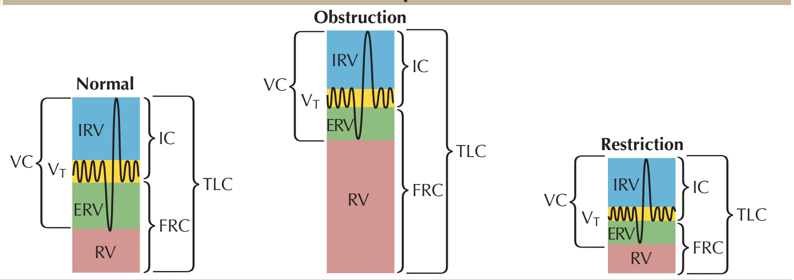
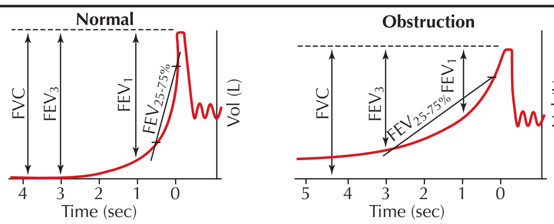
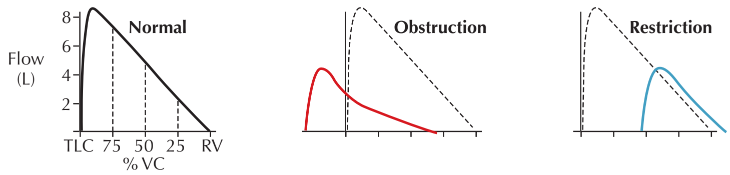
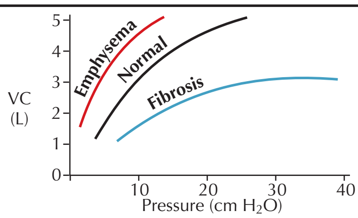
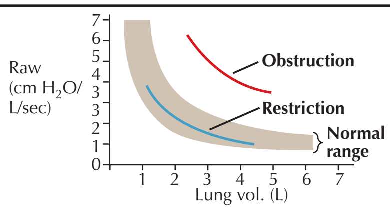
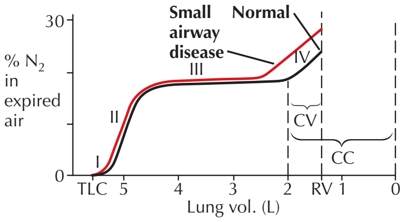
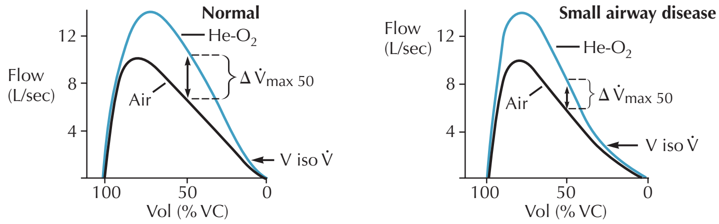
Test	Symbol	Method	Interpretation
Lung volumes and capacities Vital capacity Inspiratory capacity Expiratory reserve volume Tidal volume Functional residual capacity Residual volume Total lung capacity	 **VC** **IC** **ERV** **V_T** **FRC** **RV** **TLC**	Spirometer He dilution or body plethysmograph FRC − ERV VC + RV or FRC + IC	
Expiratory flow rates Forced expiratory volume in 1 second in 3 seconds Forced mid-expiratory flow	 **FEV_1** **FEV_3** **$FEF_{25-75\%}$ (FMF)**	Spirometer	 Normal: FEV_1/FVC, % > 70; FEV_3/FVC, % >95; $FEF_{25-75\%}$; wide range Obstruction: FEV_1/FVC, %; FEV_3/FVC, %; $FEF_{25-75\%}$ } Decreased **Restriction**: FEV_1/FVC, % and FEV_3/FVC, % usually normal; $FEF_{25-75\%}$ usually normal
Maximal expiratory flow–volume curve Maximal expiratory flow at 75, 50, 25 % of VC	**MEFV** **$\dot{V}_{max\ 75}$** **$\dot{V}_{max\ 50}$** **$\dot{V}_{max\ 25}$**	Spirometer with simultaneous recording of flow and volume and volume of integrated pneumotachograph	
Lung elasticity Static recoil pressure Static compliance	 **P_{stat}** **C_{stat}**	Pleural pressure is recorded with esophageal balloon while airflow is arrested at different lung volumes; changes in lung volume recorded with spirometer or pneumotachograph	 Static elastic recoil of lung is increased and static compliance reduced in diseases such as pulmonary fibrosis. Conversely, in emphysema, static lung compliance is increased and elastic recoil is reduced.
Airway resistance	**R_{aw}**	Body plethysmograph to determine alveolar pressure and pneumotachograph to measure airflow	 In obstructive lung disease, airway resistance is increased. If obstruction involves only small airways (<2 mm diameter), only minimal changes in overall resistance may result. In restrictive disorders, resistance is often reduced because of increased traction on intrathoracic airway walls.
Diffusing capacity	**Dco**	Low concentration of CO inhaled; expired gas analyzed for CO	Diffusing capacity is reduced when alveolar walls are destroyed and pulmonary capillaries are obliterated by emphysema and when alveolar-capillary membrane is thickened by edema, consolidation, or fibrosis.

Figure 14.10A **Pulmonary Function Tests** Tests of pulmonary function are defined and illustrated, with comparisons between values observed in normal lungs and in restrictive and obstructive disease.

Test	Symbol	Method	Interpretation	
Tests for small airway disease Closing volume Closing capacity	**CV** **CC**	Following a full inspiration of O_2 the expired lung volume from TLC to RV is plotted against the N_2 concentration		Airways in the lower lung zones close at low lung volumes and only those alveoli at top of lungs continue to empty. Because concentration of N_2 in alveoli of upper zones is higher, the slope of the curve abruptly increases (phase IV). Phase IV begins at larger lung volumes in individuals with even minor degrees of airway obstruction, increasing both CV and CC.
Maximal expiratory flow–volume curve breathing 80% He and 20% O_2	$\Delta\dot{V}_{max\ 50}$ V iso $\dot{V}$	Spirometer or pneumotachograph to record flow and volume	 During a maximal expiratory maneuver, resistance to airflow is normally due to turbulence and convective acceleration. Breathing He, which is less dense than air, lowers resistance and increases flow at all but the lowest volumes. In small airway disease, resistance to laminar flow makes up a larger portion of total resistanc and airflow is relatively independent of gas density. Increase in expiratory flow at 50% of VC while breathing He-O_2 ($\Delta\dot{V}_{max\ 50}$) will be less, and volume at which flows while breathing He-O_2 and while breathing air are identical (V iso $\dot{V}$) will be higher in patients with small airway disease than in normal individuals.	
Gas exchange			**Normal values**	**Abnormalities**
Partial pressure of O_2 in arterial blood	P_{O_2}	Arterial blood is collected anaerobically in heparinized syringe	60 to 100 mm Hg breathing room air at sea level; falls slightly with air	Hypoxemia indicative of ventilation/perfusion abnormalities, shunts, diffusion defect, alveolar hypoventilation
Partial pressure of CO_2 in arterial blood	P_{CO_2}		36 to 44 mm Hg	P_{CO_2} proportional to metabolic rate (CO_2 production) and inversely related to volume of alveolar ventilation
Arterial blood pH	pH		7.35 to 7.45 pH	Acidosis (pH <7.35) Respiratory (inadequate alveolar ventilation) Metabolic (gain of acid and/or loss of base) Alkalosis (pH >7.45) Respiratory (excessive alveolar ventilation) Metabolic (gain of base or loss of acid)
Alveolar-arterial O_2 difference	A-aD_{O_2} A-aP_{O_2}		<10 mm Hg breathing room air	Primarily reflects mismatching of ventilation and perfusion and/or shunts; may also be affected by diffusion defects
Dead space/tidal volume ratio	$\dot{V}_D/\dot{V}_T$	Determined from arterial and mixed expired P_{CO_2}	<0.3	Elevated ratio indicates wasted ventilation; i.e., that volume of gas which does not take part in gas exchange
Shunt fraction	$\dot{Q}_S/\dot{Q}_T$	Determined from P_{O_2} after a period of breathing 100% O_2	<5%	Elevation indicates increased amount of mixed venous blood entering systemic circulation without coming into contact with alveolar air, either because of shunting of blood past lungs to left side of heart or perfusion of regions of lung which are not ventilated

F. Netter M.D.

Figure 14.10B **Pulmonary Function Tests *(cont'd)***

CHAPTER 15

Oxygen and Carbon Dioxide Transport and Control of Respiration

While Fick's law describes the diffusion of gases across the alveolar-capillary membrane, many other factors are important in determining the actual content of gases in blood and in the process of gas transport. The contribution of other factors can be gleaned by considering that average oxygen consumption in humans at rest is 250 milliliters of oxygen per minute (mL O_2/min), a rate that cannot be supported simply by diffusion and delivery of dissolved gas to tissues. In fact, transport of gases is a complex process that affects other processes beyond respiration, including acid–base balance, which will also be discussed in this chapter.

TRANSPORT OF OXYGEN

The concentration of dissolved gas in a liquid is directly proportional to the partial pressure of the gas in the atmosphere to which the liquid is exposed (Henry's law) and the solubility of the gas in the solvent. For each mm Hg PO_2, only 0.003 mL O_2 will dissolve in 100 mL of blood at body temperature. Because Pa_{O_2} (PO_2 in arterial blood) is normally approximately 100 mm Hg (equilibrated with PA_{O_2}, PO_2 in alveolar air), the amount of **dissolved oxygen** in arterial blood is normally only 0.3 mL O_2/100 mL blood (Fig. 15.1). The actual measured concentration of oxygen in normal arterial blood is approximately 20.4 mL O_2/100 mL blood. What accounts for this large difference? The answer is binding of O_2 to the **hemoglobin** in red blood cells. The average concentration of hemoglobin (Hb) is 15 g/100 mL blood, and each gram (g) of hemoglobin binds 1.34 mL O_2 when fully saturated. Therefore, by far, the bulk of oxygen transport is accomplished by hemoglobin.

Oxygen Binding Capacity and Oxygen Content of Blood

The oxygen binding capacity of blood is given by the following formula:

$$O_2 \text{ binding capacity} = (1.34 \text{ mL } O_2/\text{g Hb}) \times (\text{g Hb}/100 \text{ mL blood})$$

Thus, for the normal hemoglobin concentration of 15 g/100 mL blood,

$$O_2 \text{ binding capacity} = (1.34 \text{ mL } O_2/\text{g Hb}) \times (15 \text{ g Hb}/100 \text{ mL blood}) = 20.1 \text{ mL } O_2/100 \text{ mL blood}$$

Oxygen content of blood can be calculated by the following formula:

$$O_2 \text{ content} = \% \text{ saturation} \times O_2 \text{ binding capacity} + \text{dissolved oxygen}$$

Given normal values of 15 g hemoglobin/100 mL blood and approximately 100% oxygen saturation of arterial blood at PO_2 of 100 mm Hg, oxygen content in arterial blood can be calculated as shown below:

$$\text{Arterial } O_2 \text{ content} = 100\% \times (1.34 \text{ mL } O_2/\text{g Hb}) \times (15 \text{ g Hb}/100 \text{ mL blood}) + (0.003 \text{ mL } O_2/100 \text{ mL blood/mm Hg}) \times (100 \text{ mm Hg}) = 20.4 \text{ mL } O_2/100 \text{ mL blood}$$

Arteriovenous Oxygen Gradient and Oxygen Consumption

Compared with this arterial O_2 content (20.4 mL O_2/100 mL blood), in normal venous blood, in which PO_2 is approximately 40 mm Hg and hemoglobin is 75% saturated,

$$\text{Venous } O_2 \text{ content} = 75\% \times (1.34 \text{ mL } O_2/\text{g Hb}) \times (15 \text{ g Hb}/100 \text{ mL blood}) + (0.003 \text{ mL } O_2/100 \text{ mL blood/mm Hg}) \times (40 \text{ mm Hg}) = 15.2 \text{ mL } O_2/100 \text{ mL blood}$$

Each deciliter of blood delivers approximately 5 mL of O_2 to the tissues as it passes through the systemic circulation, because the arteriovenous difference in oxygen content of blood is 5 mL. Based on this arteriovenous difference in

Oxygen binding capacity is the maximum amount of oxygen that can be bound to hemoglobin in blood (at 100% saturation). The actual oxygen content of blood depends on the degree of saturation of hemoglobin. Dissolved oxygen is a minor portion of the oxygen content of blood; in a person breathing normal air, it is only 0.3 mL/100 mL blood, compared to the total content of oxygen of 20.4 mL/100 mL blood at 100% saturation.

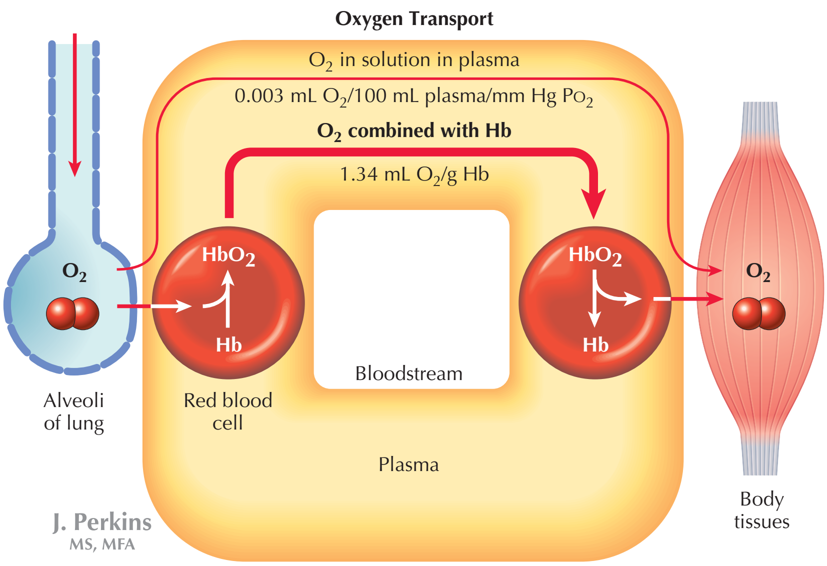


Figure 15.1 Oxygen Transport Oxygen diffuses into the blood flowing through alveolar capillaries and is transported to tissues, where it diffuses out of the blood along its concentration gradient. Transport of oxygen in the blood is mainly in the form of oxygen combined with hemoglobin, with only a minor portion carried in the form of dissolved oxygen.

oxygen content of blood and cardiac output, oxygen consumption can be estimated. Thus, at rest,

$$O_2 \text{ consumption} = [a - v]_{O_2} \times \text{cardiac output} = (5 \text{ mL } O_2/100 \text{ mL}) \times 5000 \text{ mL/min} = 250 \text{ mL } O_2/\text{min}$$

At the usual respiratory quotient (CO_2 production/O_2 consumption) of 0.8, CO_2 production is 200 mL/min.

THE OXYHEMOGLOBIN DISSOCIATION CURVE

As discussed, at PO_2 of 100 mm Hg, hemoglobin is nearly 100% saturated with oxygen (actually, 97.5%), and at 40 mm Hg, hemoglobin is 75% saturated. The **oxyhemoglobin dissociation curve** (Fig. 15.2) describes the relationship between oxygen saturation of blood (SO_2) and PO_2. The sigmoidal shape of the curve is due to cooperative binding of hemoglobin. Each molecule of hemoglobin is capable of binding four oxygen molecules; when one molecule binds to an oxygen binding site, the other three sites bind oxygen more readily, causing the middle portion of the curve to be steep. Thus, exposure of blood to the high PO_2 in the respiratory zone of the lung results in binding of a substantial amount of oxygen. Note also that due to the sigmoidal shape, PA_{O_2} can fall substantially without greatly affecting the degree of saturation of hemoglobin (at PO_2 of 80 mm Hg, SO_2 is still above 95%). On the other hand, as PO_2 falls in blood coursing through systemic capillaries, typically to 40 mm Hg, blood PO_2 is on the steep portion of the curve, which facilitates delivery of oxygen to the tissues. Dissociation of one molecule of oxygen from a molecule of hemoglobin makes dissociation of the other bound oxygen molecules from that molecule more likely.

To summarize:

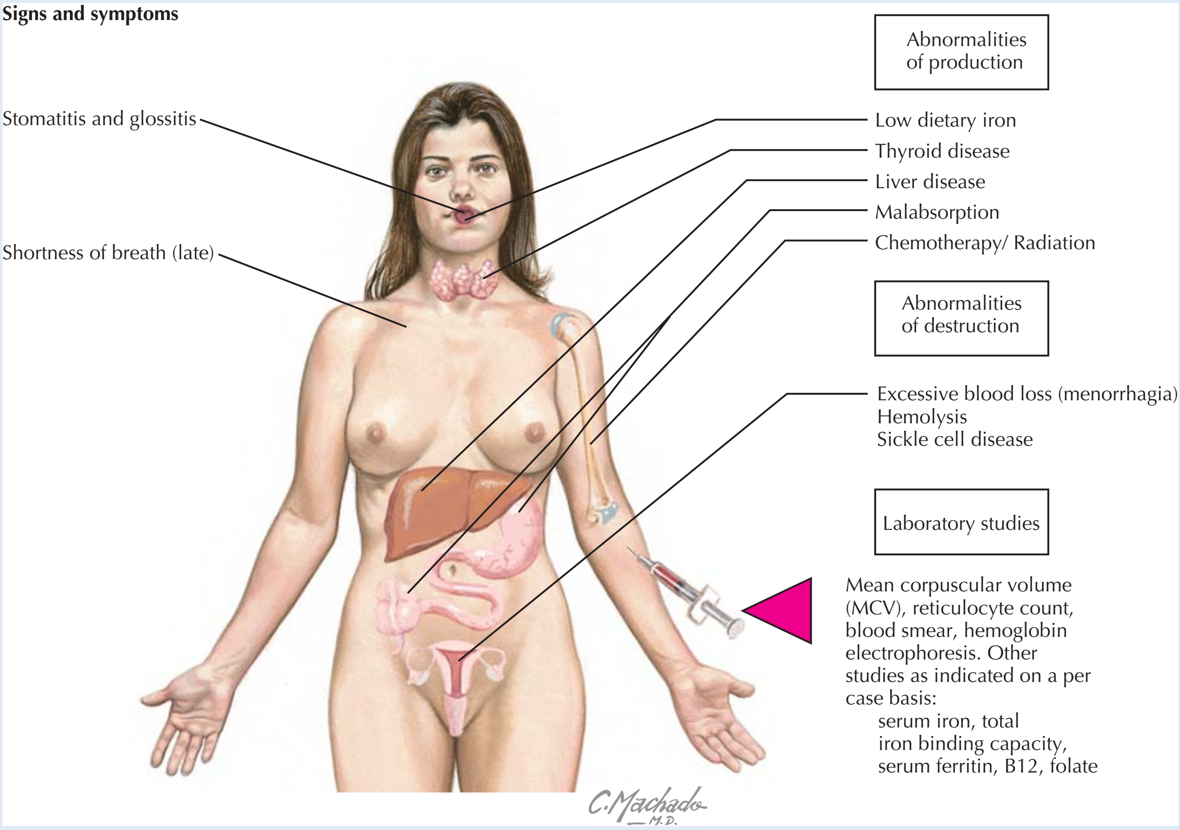
- Cooperative binding of oxygen to four binding sites per hemoglobin molecule is responsible for the sigmoidal shape of the oxyhemoglobin dissociation curve.
- In the lungs, blood will become fully saturated with oxygen over a wide range of PO_2, due to the flat, upper portion of the oxyhemoglobin dissociation curve.
- At the lower PO_2 levels in tissue capillaries, small changes in PO_2 result in dissociation of relatively large amounts of oxygen, due to the steep middle portion of the curve, facilitating oxygen delivery to the tissues.

Factors Affecting the Oxyhemoglobin Dissociation Curve

In addition to these qualities, another important characteristic of the oxyhemoglobin dissociation curve is that it is shifted to the right under conditions of increased PCO_2, low pH, and high temperature (see Fig. 15.2). These conditions occur locally during tissue hypoxia and increased metabolism (for example, during exercise), and the rightward shift of the curve results in decreased hemoglobin affinity for oxygen, and thus enhances delivery of oxygen to tissues. The metabolite of red blood cell glycolysis, **2,3-diphosphoglycerate** (**2,3-DPG;** also known as 2,3-bisphophoglycerate), also shifts the curve to the right and is elevated during hypoxia.

CLINICAL CORRELATE
Anemia

Anemia is the state of low hemoglobin or red blood cell count in blood, resulting in reduced oxygen binding capacity. It may be the result of blood loss, insufficient erythropoiesis (reduced red blood cell production), or hemolysis (red blood cell destruction). Iron deficiency may also cause anemia in menstruating women. Hypoxia associated with anemia cannot be corrected by administration of oxygen, because only the minor proportion of oxygen that is dissolved in arterial blood will be raised by elevating alveolar oxygen concentration, and oxyhemoglobin (oxygen bound to hemoglobin) will remain low due to the hemoglobin deficiency. Although mild anemia can be treated by a variety of interventions that address the primary cause of the anemia, severe anemia requiring immediate medical attention must be treated by blood transfusion.



Anemia Destruction of red blood cells, bleeding, or abnormalities of red blood cell production are causes of anemia. Laboratory studies are useful in differential diagnosis of the cause.

Oxygen delivery to the fetus involves transfer of oxygen from maternal blood to fetal blood across the placenta. Fetal blood contains a form of hemoglobin known as fetal hemoglobin (hemoglobin F) that has higher affinity for oxygen than adult hemoglobin (hemoglobin A), facilitating transfer of oxygen from maternal to fetal blood. Thus, the oxyhemoglobin dissociation curve for hemoglobin F is shifted to the left, compared with the dissociation curve for hemoglobin A. At a given PO_2, oxygen saturation of hemoglobin F will be higher than that of hemoglobin A. Hemoglobin F is replaced by hemoglobin A within the first 3 months of life.

TRANSPORT OF CARBON DIOXIDE

The concentration of carbon dioxide is highest in the mitochondria, where it is produced during cellular respiration. From there, it diffuses to the interstitium and eventually into the blood, which transports it to the alveoli. Within the blood, CO_2 is transported in three forms (Fig. 15.3):

- Approximately 7% of CO_2 in blood is **dissolved CO_2.** Because solubility of CO_2 in plasma is relatively high (20 times the solubility of O_2), the dissolved form of CO_2 has a significant role in its transport.

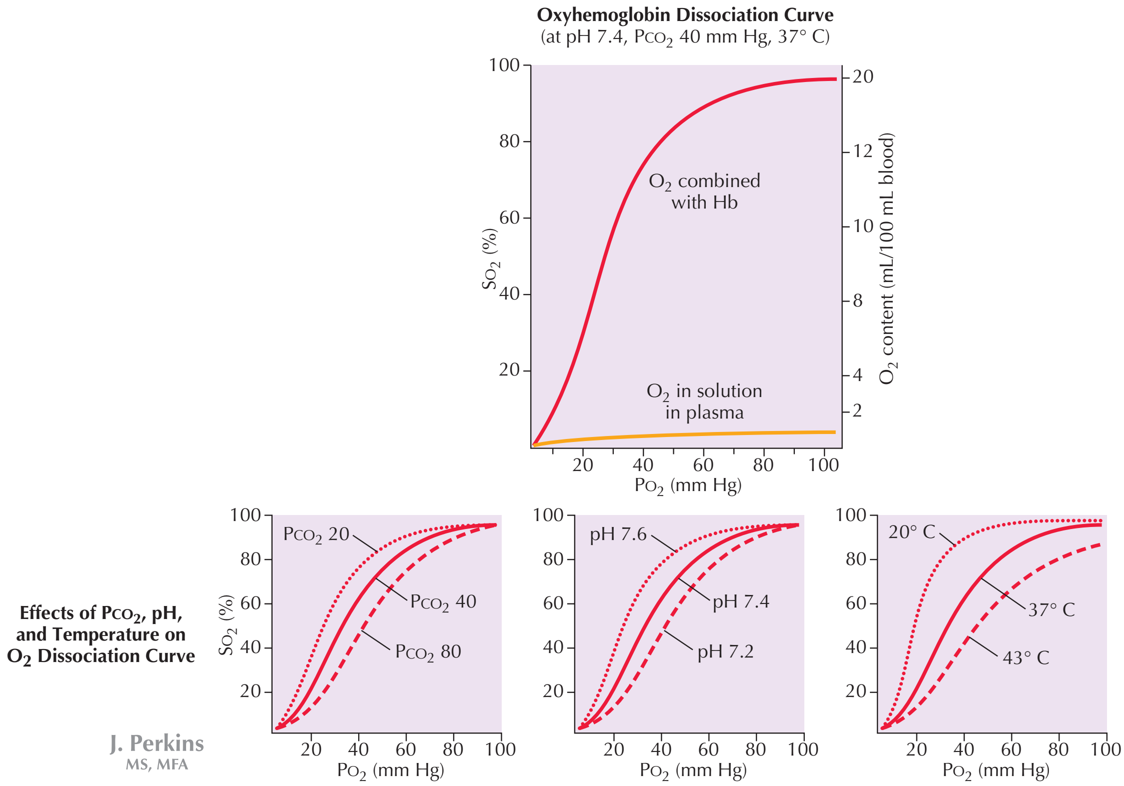


Figure 15.2 Oxyhemoglobin Dissociation Curves The oxyhemoglobin binding curve describes the association between partial pressure of oxygen and the degree of oxygen saturation of hemoglobin. Saturation of hemoglobin is nearly 100% (97.5%) when P_{O_2} is 100 mm Hg. The sigmoidal shape of this curve results in a high degree of oxygen saturation of blood after passing through alveolar capillaries and significant dissociation of oxygen from hemoglobin at the P_{O_2} levels to which blood is exposed as it perfuses systemic capillaries. The values given for oxygen content on the graph are those expected at the normal blood hemoglobin concentration of 15 g/100 mL blood. Note that the amount of dissolved hemoglobin in blood is very low over a wide range of P_{O_2}. High P_{CO_2}, low pH, and high temperature shift the oxyhemoglobin dissociation curve to the right, which promotes oxygen dissociation from hemoglobin in capillaries supplying actively metabolizing tissues.

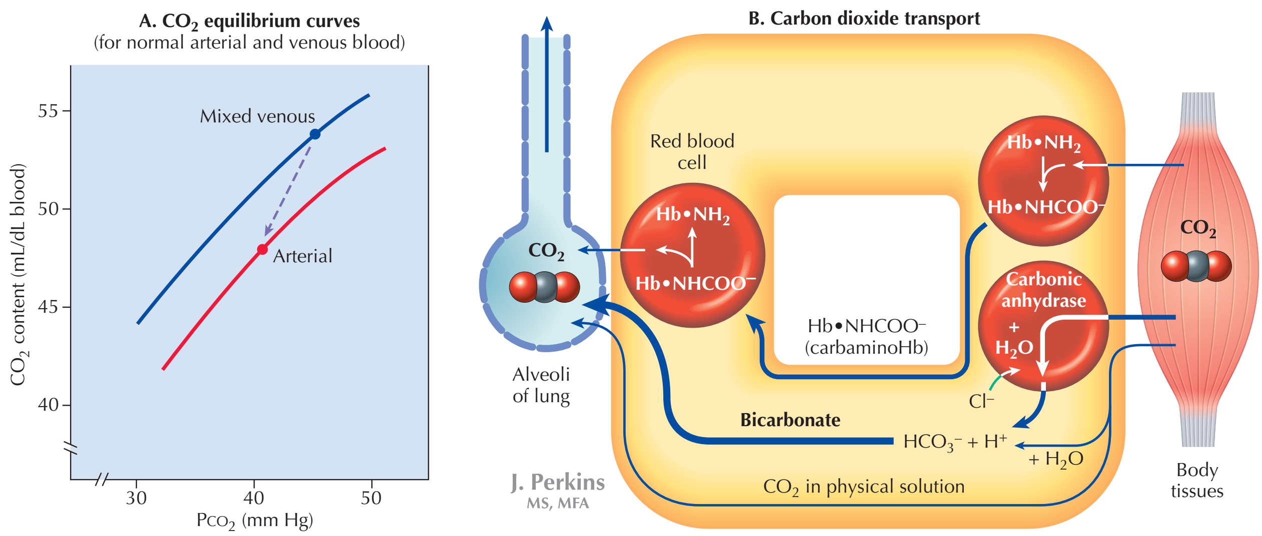
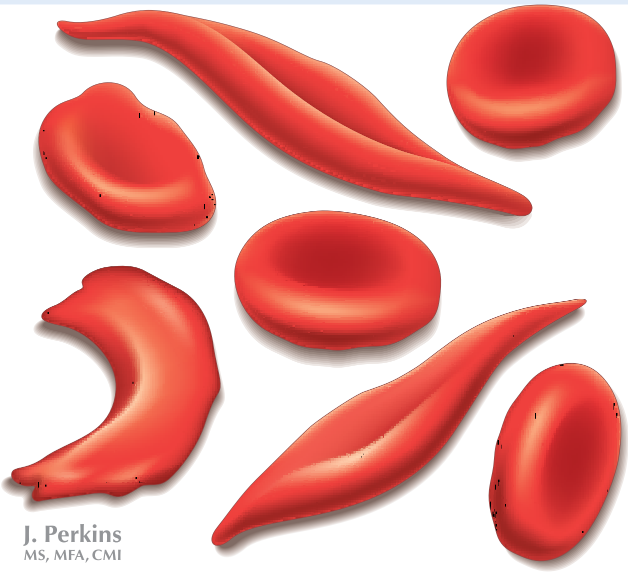


Figure 15.3 Carbon Dioxide Transport Carbon dioxide is transported in blood as bicarbonate anion (approximately 70%), carbaminohemoglobin (approximately 23%), and dissolved CO_2 (approximately 7%). The CO_2 equilibrium (dissociation) curve *(A)* is steep and linear, unlike the oxyhemoglobin dissociation curve, accounting for the relatively small difference in P_{CO_2} between arterial and venous blood (40 mm Hg vs. 45 mm Hg). Note also that the dissociation curve is shifted to the left when hemoglobin is in the form of deoxyhemoglobin, as in venous blood (the Haldane effect).

CLINICAL CORRELATE
Sickle Cell Disease

Patients suffering from sickle cell disease have a variant form of hemoglobin known as hemoglobin S. The allele causing this disease is recessive and is most common in people of sub-Saharan African origin. Hemoglobin S has the tendency to polymerize when it is deoxygenated, causing red blood cells to assume a sickle-like shape. In a “sickle cell crisis,” red blood cells lodge in the microcirculation, causing painful ischemia and infarction of tissue. Sickle cell disease confers resistance to malaria, a parasite attacking red blood cells, and is believed to have evolved and persisted in sub-Saharan Africa due to evolutionary advantage associated with malaria resistance.



Sickled Red Blood Cells

- Up to 23% of CO_2 may be combined with protein, including hemoglobin (as **carbaminohemoglobin,** which gives venous blood its bluish tinge). CO_2 binds to terminal amino groups of blood proteins.
- About 70% of CO_2 in blood is carried in the form of **bicarbonate anion** (HCO_3^-).

Carbon Dioxide Transport in the Form of Bicarbonate Ion

The bulk of CO_2 is transported in the form of HCO_3^- within red blood cells (see Fig. 15.3). CO_2 dissolved in blood reacts with H_2O to form carbonic acid (H_2CO_3), which dissociates to form H^+ and HCO_3^-:

$$CO_2 + H_2O \leftrightarrow H_2CO_3 \leftrightarrow H^+ + HCO_3^-$$

This reaction, which is normally slow, is catalyzed by the enzyme **carbonic anhydrase** in red blood cells. As HCO_3^- is formed, it diffuses out of the red blood cell while Cl^- diffuses into the cell to maintain electrochemical equilibrium. This process is known as the **chloride shift**. Most of the H^+ formed is buffered within the red blood cell by binding to hemoglobin. The reaction forming H_2CO_3 is driven forward in capillary blood, as CO_2 diffuses from tissues into the blood. At the lungs, the reverse reaction occurs, as CO_2 is breathed off.

In normal venous blood, the partial pressures of carbon dioxide and oxygen are similar (approximately 45 mm Hg and 40 mm Hg, respectively). According to Henry’s law, the concentration of dissolved gas in a solution is directly proportional to its partial pressure. Although this law applies to both oxygen and carbon dioxide, because the solubility of carbon dioxide is 20 times that of oxygen, at similar partial pressures, much more carbon dioxide is present in blood in the form of dissolved gas than is the case for oxygen.

Although the approximate values given earlier for CO_2 transport as dissolved gas, carbaminohemoglobin, and bicarbonate anion are prevalent in the literature and textbooks, the value for carbaminohemoglobin may be significantly overstated, because the original studies of CO_2 carriage in blood were performed in the absence of 2,3-DPG, which binds with higher affinity to hemoglobin than CO_2.

The Haldane Effect

Unlike the sigmoidal oxyhemoglobin dissociation curve, the dissociation curve for CO_2 in blood is linear (see Fig. 15.3). However, it is shifted to the left when hemoglobin is in the form of deoxyhemoglobin, as in venous blood. This is known as the **Haldane effect.** As a result of the Haldane effect, as hemoglobin is deoxygenated in systemic capillaries, its affinity for CO_2 is increased, facilitating CO_2 transport. Binding affinity for H^+ (generated in red blood cells along with HCO_3^-) is also increased. In the pulmonary circulation, as hemoglobin is oxygenated, its affinity for CO_2 is reduced, and as a result, transfer of CO_2 from blood to alveolar air is facilitated.

CARBON DIOXIDE TRANSPORT AND ACID–BASE BALANCE

Normal blood pH is approximately 7.4 and is regulated within a tight range (7.35 to 7.45) by renal and respiratory mechanisms and by various other buffering systems (Fig. 15.4 and see Chapter 20). Significant deviation of pH from the normal range is incompatible with life, as protein structure is affected and enzymatic function is disturbed. CO_2 transport plays an important role in maintaining **acid–base equilibrium.** The **“acid load”** of the body consists of volatile acid, which is CO_2 in its various forms, and nonvolatile acids such as lactic acid and amino acids. Nonvolatile acids are buffered by intracellular and extracellular mechanisms; the bicarbonate buffering system of extracellular fluids including blood are